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AI Engineer · LLM Systems · Agentic Pipelines · AWS Certified Cloud Practitioner

Kolkata | <https://github.com/Arkoprobho-Mitra> | AWS Cloud Practitioner | Springer Research Author

PROFESSIONAL SUMMARY

AI Engineer with 5+ years at Infosys, with a track record of shipping production GenAI and ML systems at Infosys Topaz. Currently leading sub-teams across banking RAG, insurance claims intelligence, and multi-agent procurement automation. Background in operational data systems such as PL/SQL, Elasticsearch, QlikSense, and ServiceNow, across large-scale logistics workflows. B.E. from Jadavpur University, AWS Certified Cloud Practitioner, and co-author of a Springer-published research paper on multilingual OCR. Builds AI systems engineered for production — cost-optimised, secure, and observable from day one.

WORK EXPERIENCE

AI Engineer / Sub-team Lead

Nov 2023 – Present

Infosys Topaz (Applied AI Practice) · Kolkata, India

Project 3 · Multi-Agent Procurement Copilot (Aug 2025 – Present)

- ▶ Leading a team of 3 AI Engineers to architect and ship a LangGraph-orchestrated multi-agent system for a global manufacturing enterprise, delivering a 35% reduction in procurement cycle time and cutting RFQ turnaround from 3 days to 4 hours.
- ▶ Architected a 5-agent graph (Planner, Supplier, Contract, Compliance, Synthesis) integrating SAP Ariba, ServiceNow, and SharePoint REST APIs via LangChain tool definitions.
- ▶ Implemented vector-backed contract RAG on Azure AI Search with metadata access-control filtering; blue-green index deployments with canary health checks eliminate silent data-loss incidents.
- ▶ Architected enterprise-grade security posture: zero-secret agent design via Azure Key Vault and Managed Identity; all write operations gated behind HITL approval nodes; immutable audit logging to Azure Blob Storage for GDPR/DPA compliance.
- ▶ Evaluated system across three levels: tool selection accuracy (> 90%), workflow completion rate (> 95%), and RAGAS faithfulness + SME scoring per sprint.

Project 2 · Insurance Claims Intelligence Platform (Nov 2024 – Sep 2025)

- ▶ Spearheaded a team of 4 AI Engineers to design and deploy an end-to-end claims intelligence platform for a global insurer, achieving a 40% reduction in processing time across OCR, field extraction, fraud detection, and HITL routing.
- ▶ Deployed Azure Document Intelligence + LayoutLMv2 classifier (94% accuracy on 15 K labelled pages) for document-type routing across 12 classes (FNOL forms, medical records, police reports, etc.).
- ▶ Built XGBoost fraud-detection model (800K claims, 3% fraud prevalence) with SHAP explainability for audit-readiness; handled class imbalance via `scale_pos_weight` rather than SMOTE.
- ▶ Integrated Evidently AI drift monitoring triggering automated MLflow retraining; moving from 5-day manual SLA to automated overnight retraining when PSI exceeds threshold.

Project 1 · Enterprise Knowledge Assistant — Global Banking (Nov 2023 – Dec 2024)

- ▶ Sub-team Lead (2 engineers): delivered a 2-phase internal GenAI assistant over a 2.5M-document corpus for a North American investment bank, reducing compliance query time by **60%** at **30,000+** monthly active users.
- ▶ Phase 1: Custom Python orchestration over Azure Cognitive Search (BM25 + Semantic Ranker), Azure Form Recognizer OCR, and GPT-3.5-Turbo on Azure for answer synthesis.
- ▶ Phase 2 (Topaz): Migrated to hybrid vector + BM25 retrieval (Reciprocal Rank Fusion); +14% precision on regulation-code queries. Upgraded to GPT-4; introduced LangChain post enterprise security review.
- ▶ Enforced role-based document access via pre-retrieval metadata filters on Azure AD group memberships; model never received restricted content.
- ▶ Built 200-query golden eval dataset with compliance SME; embedded evaluation gates in CI/CD; adopted RAGAS in Phase 2 for automated scoring.

Systems Engineer — Data & Logistics

Aug 2021 – Oct 2023

Infosys (Logistics & Data Systems Practice) · Remote / India

- Maintained and optimised PL/SQL stored procedures, queries, and data pipelines supporting daily logistics operations across large-scale shipping and supply chain workflows.
- Administered Elasticsearch indices for real-time shipment tracking and operational search; tuned mappings and queries for throughput and latency.
- Built and maintained QlikSense dashboards surfacing operational KPIs for logistics and operations teams.
- Managed ServiceNow workflows for incident and change management, integrating with client legacy systems and internal databases.
- Collaborated with cross-functional teams on client system integrations and data migration tasks, supporting scalable data architecture across the logistics practice.

PERSONAL PROJECTS

Agentic Job Application Pipeline

2024

- LangGraph multi-agent system that scrapes job listings, scores them with an LLM, drafts tailored cover letters, and tracks application state — full agentic automation of the job search workflow.

Full-stack E-Commerce Platform with RAG Assistant

2024

- Microservices architecture: React 19 + Redux, Spring Boot 3 (Java 17), FastAPI + asyncpg, Node.js semantic search (@xenova/transformers). RoBERTa + XGBoost review sentiment pipeline; Gemini 1.5 Flash RAG assistant; distributed JWT auth across all 3 services.

LangGraph Fact-Checker with Human-in-the-Loop

2024

- Multi-node LangGraph graph with web search tool, claim verification agent, and HITL approval step; Langfuse observability integration for trace-level evaluation.

TECHNICAL SKILLS

AI / LLM Frameworks: LangGraph, LangChain, LangFuse, FAISS, RAGAS, CrewAI, MCP, Pydantic AI, Anthropic SDK, OpenAI SDK, LiteLLM

Machine Learning: pandas, XGBoost, LightGBM, scikit-learn, SHAP, LayoutLMv2, RoBERTa, SpaCy NER, Evidently AI, MLflow, Azure ML

Backend: Python, FastAPI, asyncio, asyncpg, Spring Boot 3 (Java 17), Node.js, APScheduler, SQLite, PostgreSQL

Frontend: React 19, Redux Toolkit, Vite, Tailwind CSS, JavaScript, Google Apps Script

Database & Search: PL/SQL, Elasticsearch, Azure AI Search (Cognitive Search), PostgreSQL, SQLite

Cloud & Infrastructure: AWS EC2, Azure OpenAI, Azure Document Intelligence, Azure Key Vault, Azure Blob Storage, Docker, Vercel, GitHub Actions

Enterprise Tooling: QlikSense, ServiceNow, SAP Ariba REST API, SharePoint REST API

EDUCATION

Bachelor of Engineering (B.E.),

2017 – 2021

Jadavpur University, Kolkata, India

CERTIFICATIONS

AWS Certified Cloud Practitioner · Amazon Web Services

RESEARCH PUBLICATION

How to handle bi/tri-lingual Indic texts in a single image? A new dataset of natural scene and born-digital images

Multimedia Tools and Applications, Springer · Vol. 81, pp. 15367–15394 (2022)

Department of CSE, Jadavpur University

DOI: <https://doi.org/10.1007/s11042-022-12596-7>

Research scope: constructed and published a novel multilingual dataset of natural scene and born-digital images containing Indic script combinations, advancing OCR and scene-text detection research for low-resource scripts.